

Bazı Önemli İşaret Tipleri **(Some Important Signal Types)**

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Textbook: Linear Systems and Signals, B.P. Lathi

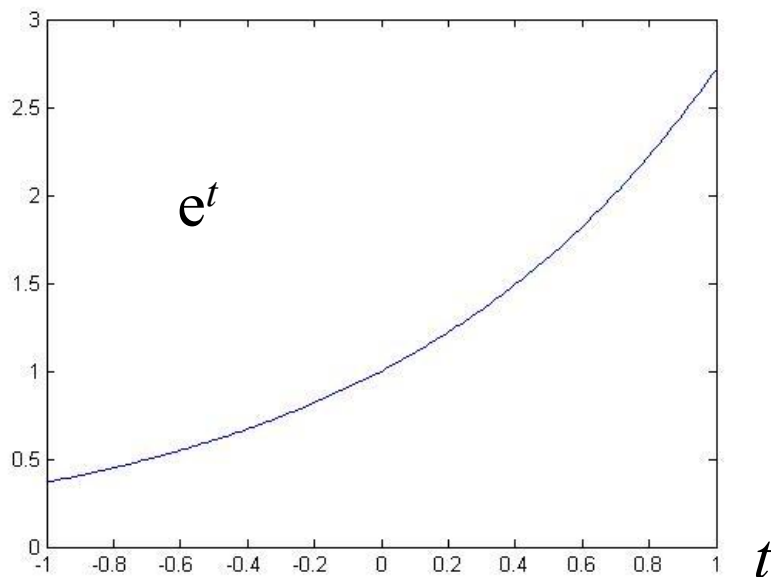
This slides are modified from Linear Systems and Signals (EE313) course at the University of Texas at Austin, with Courtesy of Prof. Brian L. Evans

Signal Representation

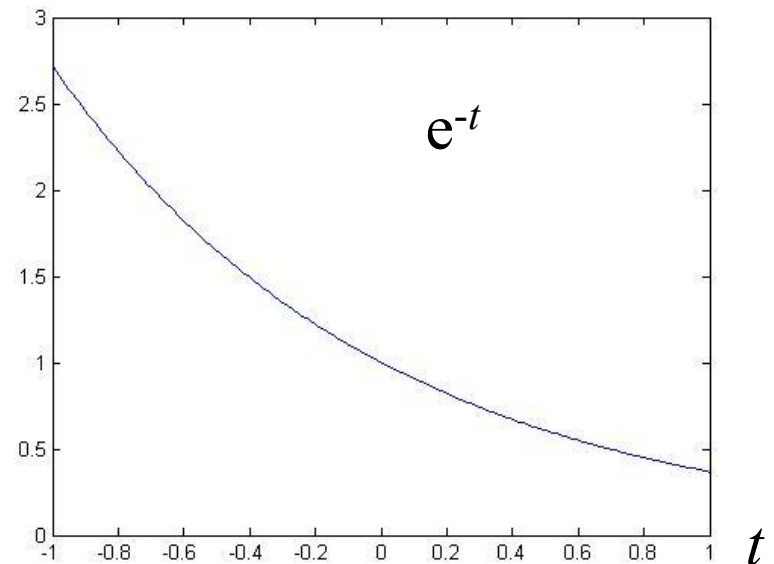
- A function, e.g. $\sin(t)$ in continuous-time or $\sin(2 \pi n / 10)$ in discrete-time, useful in *analysis*
- A sequence of numbers, e.g. $\{1,2,3,2,1\}$ which is a sampled triangle function, useful in *simulation*
- A collection of properties, e.g. even symmetric about origin, useful in *reasoning* about behavior
- A piecewise representation, e.g. $u(t) = \begin{cases} 1 & \text{for } t > 0 \\ \frac{1}{2} & \text{for } t = 0 \\ 0 & \text{for } t < 0 \end{cases}$
- A functional, e.g. the Dirac delta functional $\delta(t)$ $u[n] = \begin{cases} 1 & \text{for } n \geq 0 \\ 0 & \text{otherwise} \end{cases}$

Exponential Signals

- **Solutions to linear constant-coefficient differential equations, and hence, very common**



```
t = -1 : 0.01 : 1;  
e1 = exp(t);  
plot(t, e1)
```



```
t = -1 : 0.01 : 1;  
e2 = exp(-t);  
plot(t, e2)
```

Exponential Signal Properties

- **Real-valued exponential signals**

Amplitude values are always non-negative

Might decay or not as t goes to infinity

$$\lim_{t \rightarrow \infty} e^{at} = \begin{cases} 0 & \text{if } a < 0 \\ 1 & \text{if } a = 0 \\ \infty & \text{if } a > 0 \end{cases}$$

- **Complex-valued exponential signals**

$$e^{j\theta} = \cos(\theta) + j \sin(\theta)$$

$$e^{-j\theta} = \cos(\theta) - j \sin(\theta)$$



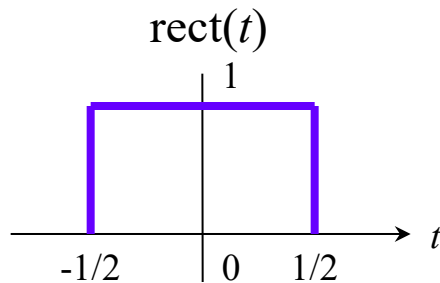
$$e^{j\theta} + e^{-j\theta} = 2 \cos(\theta)$$

$$e^{j\theta} - e^{-j\theta} = 2j \sin(\theta)$$

We'll need these properties throughout the semester

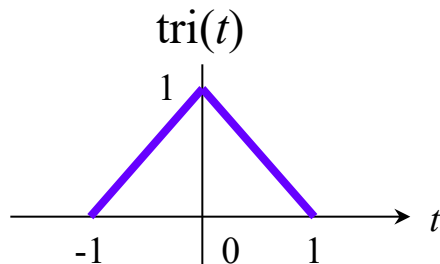
Piecewise Functions

- Unit area rectangular pulse



$$\text{rect}(t) = \begin{cases} 1 & |t| < \frac{1}{2} \\ \frac{1}{2} & |t| = \frac{1}{2} \\ 0 & |t| > \frac{1}{2} \end{cases}$$

- What does $\text{rect}(x / a)$ look like?
- Unit triangle function



$$\text{tri}(t) = \begin{cases} 1 - |t| & |t| < 1 \\ 0 & |t| \geq 1 \end{cases}$$

Math commands

`rectpuls(t)`

`tripuls(0.5*t)`

Both functions are even symmetric about origin.

Dirac Delta Functional

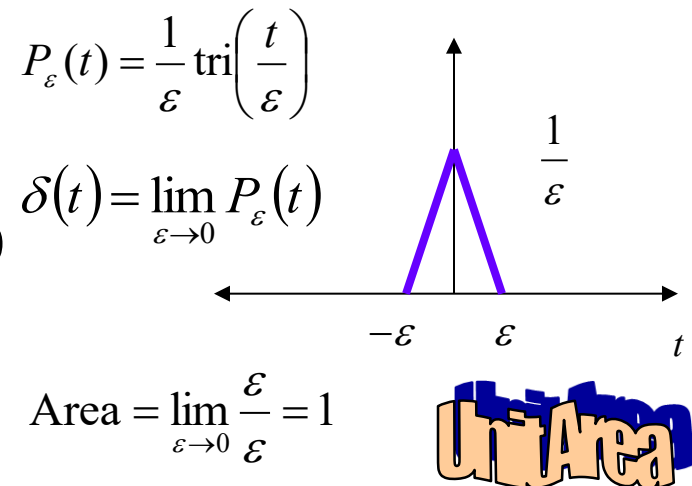
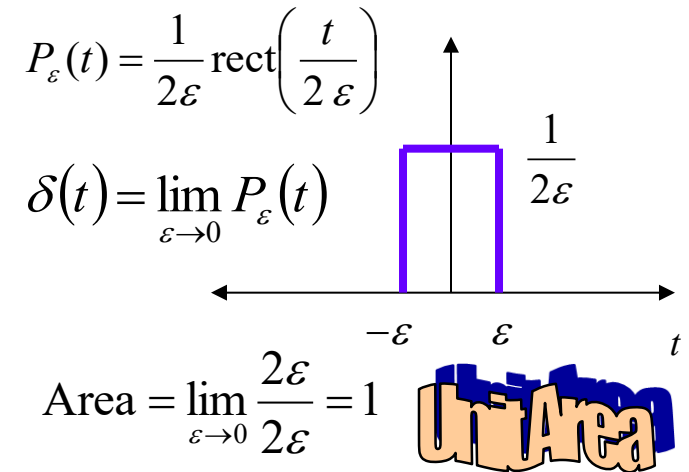
- Mathematical idealism for an instantaneous event
- Dirac delta as generalized function (a.k.a. functional)

Unit area: $\int_{-\infty}^{\infty} \delta(t) dt = 1$

Sifting $\int_{-\infty}^{\infty} g(t) \delta(t) dt = g(0)$
 provided $g(t)$ is **defined** at $t = 0$

Scaling: $\int_{-\infty}^{\infty} \delta(at) dt = \frac{1}{|a|}$ if $a \neq 0$

- Note that $\delta(0)$ is undefined



Dirac Delta Functional

- Generalized sifting, assuming that $a > 0$

$$\int_{-a}^a \delta(t-T) dt = \begin{cases} 1 & \text{if } -a < T < a \\ 0 & \text{if } T < -a \text{ or } T > a \end{cases}$$

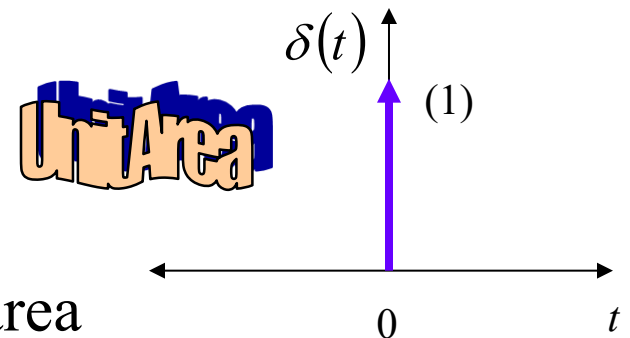
- By convention, plot Dirac delta as arrow at origin

Undefined amplitude at origin

Denote area at origin as (area)

Height of arrow is irrelevant

Direction of arrow indicates sign of area



- Simplify Dirac delta terms only under integration

Dirac Delta Functional

- We can simplify $\delta(t)$ under integration

$$\int_{-\infty}^{\infty} \phi(t) \delta(t) dt = \phi(0)$$

- What about?

$$\int_{-\infty}^{-1} \phi(t) \delta(t) dt = ?$$

Answer: 0

- What about?

$$\int_{-\infty}^{\infty} \phi(t) \delta(t - T) dt = ?$$

By substitution of variables,

$$\int_{-\infty}^{\infty} \phi(t + T) \delta(t) dt = \phi(T)$$

- Other examples

$$\int_{-\infty}^{\infty} \delta(t) e^{-j\omega t} dt = 1$$

$$\int_{-\infty}^{\infty} \delta(t - 2) \cos\left(\frac{\pi t}{4}\right) dt = 0$$

$$\int_{-\infty}^{\infty} e^{-2(x-t)} \delta(2-t) dt = e^{-2(x-2)}$$

- What about at origin?

$$\int_{-\infty}^0 \delta(t) dt = ?$$

$$\int_{-\infty}^{0^-} \delta(t) dt = 0$$

$$\int_{-\infty}^{0^+} \delta(t) dt = 1$$

Before Impulse

After Impulse

Unit Step Function

- Models event that turns on and stays on

- Definition

$$u(t) = \int_{-\infty}^t \delta(\tau) d\tau = \begin{cases} 0 & t < 0 \\ ? & t = 0 \\ 1 & t > 0 \end{cases} \quad \longleftrightarrow \quad \frac{du}{dt} = \delta(t)$$

- What happens at the origin for $u(t)$?

$u(0^-) = 0$ and $u(0^+) = 1$, but $u(0)$ can take any value

Textbook uses $u(0) = 1/2$ to average left and right hand limits

Impulse invariance filter design uses $u(0) = 1/2$

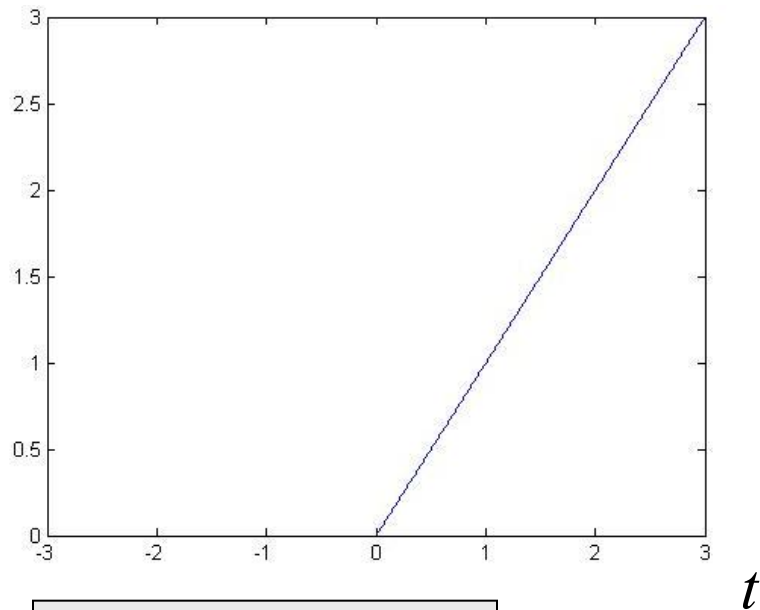
L. B. Jackson, "A correction to impulse invariance," *IEEE Signal Processing Letters*, vol. 7, no. 10, Oct. 2000, pp. 273-275.

Math command `stepfun(t,0)` defines $u(0) = 1$

Other Important Functions

- **Ramp**

$$\text{ramp}(t) = t u(t)$$

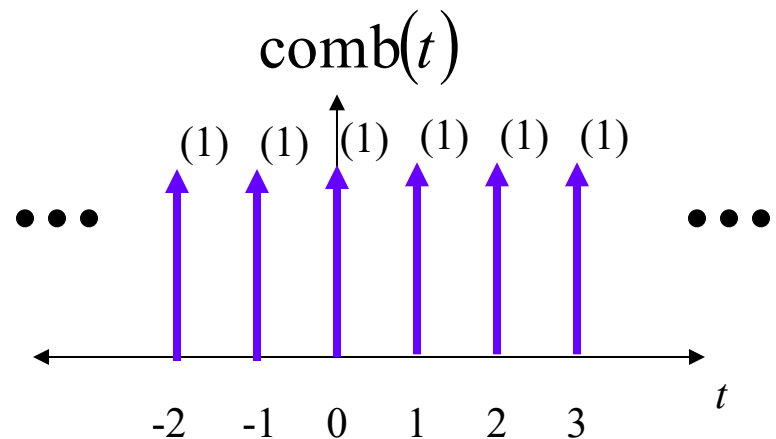


```
t = -3 : 0.01 : 3;  
r = t .* stepfun(t,0);  
plot(t, r)
```

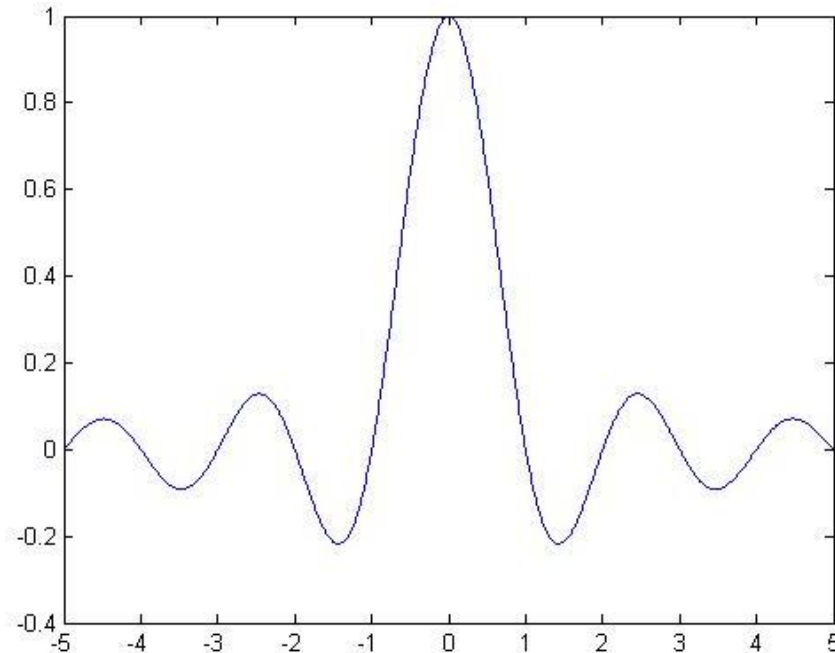
- **Unit comb**

Impulse train

$$\text{comb}(t) = \sum_{n=-\infty}^{\infty} \delta(t-n)$$



Sinc Function



```
t = -5 : 0.01 : 5;  
s = sinc(t);  
plot(t, s)
```

$$\text{sinc}(t) = \frac{\sin(\pi t)}{\pi t}$$

How to compute $\text{sinc}(0)$?

As $t \rightarrow 0$, numerator and denominator are both going to 0. How to handle it?

Even symmetric about origin

Zero crossings at $t = \pm 1, \pm 2, \pm 3, \dots$

Amplitude decreases proportionally to $1/t$

Sampling

- Many signals originate as continuous-time signals, e.g. voice or conventional music
- Sample continuous-time signal at equally-spaced points in time to obtain discrete-time signal

$$y[n] = y(n T_s)$$

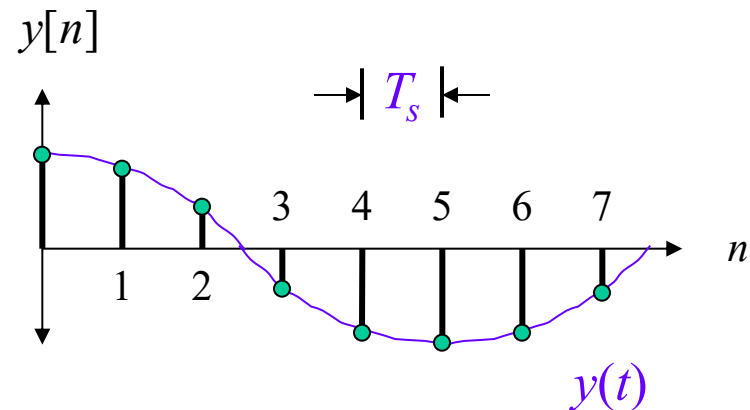
$$n \in \{\dots, -2, -1, 0, 1, 2, \dots\}$$

T_s is sampling period

- **Example**

$$y(t) = A \cos(2 \pi f_0 t + \phi)$$

$$y[n] = y(n T_s) = A \cos(2 \pi f_0 T_s n + \phi)$$



Audio CD Samples at 44.1 kHz

- Human hearing is from about 20 Hz to 20 kHz
- *Sampling theorem* (covered at mid-semester): sample continuous-time signal at rate of more than twice highest frequency in signal
- **Analog-to-digital conversion for audio CD**

First, apply a *filter* to pass frequencies up to 20 kHz (called a *lowpass* filter) and reject high frequencies.

Lowpass filter needs 10% of cutoff frequency to roll off to zero (filter can reject frequencies above 22 kHz)

Second, sample at 44.1 kHz captures analog frequencies of up to but not including 22.05 kHz

Third, quantize to 16 bits per sample

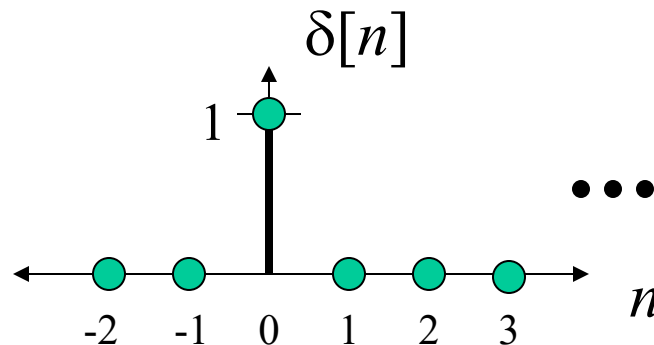
Discrete-Time Impulse and Step

- **Impulse *function***

$$\delta[n] = \begin{cases} 1 & n = 0 \\ 0 & n \neq 0 \end{cases}$$

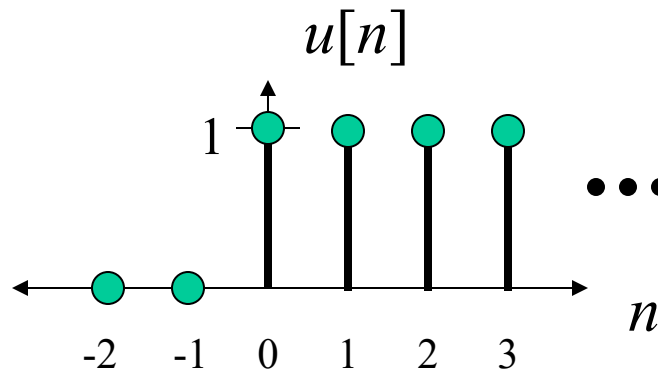
Also called Kronecker Delta

Even symmetric about origin



- **Unit step (unit sequence)**

$$u[n] = \begin{cases} 1 & n \geq 0 \\ 0 & n < 0 \end{cases}$$



```
n = -2 : 3;  
u = stepfun(n,0);  
stem(n, u);
```

Discrete-Time Sinusoidal Signals

Sinusoidal signal in continuous time

$$y(t) = A \cos(2\pi f_0 t + \phi)$$

Sample using sampling period T_s

$$y[n] = y(n T_s) = A \cos(2\pi f_0 T_s n + \phi)$$

Substitute $T_s = 1 / f_s$,
 f_s is sampling rate,

$$y[n] = A \cos\left(2\pi \frac{f_0}{f_s} n + \phi\right)$$

- Discrete-time frequency

$$\Omega_0 = 2\pi \frac{f_0}{f_s} \text{ in rad/sample}$$

Given integers N and L with common factors removed, discrete-time

sinusoid has period L if $\frac{f_0}{f_s} = \frac{N}{L}$

- Example: singing a tone during cell phone call

$$\frac{f_0}{f_s} = \frac{1000 \text{ Hz}}{8000 \text{ Hz}} = \frac{1}{8}$$